JEE Mains 2019 Chapter wise Question Bank

Sequences and Series – Questions

Q1

If a, b and c be three distinct real numbers in G.P. and $a + b + c = xb$, then x cannot be:

(1) -2 (2) -3

(3) 4 (4) 2

9 Jan Morning

Q2

Let $a_1, a_2, \ldots, a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{(2i-1)}$.

If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to:

(1) 52 (2) 57

(3) 47 (4) 42

9 Jan Morning

Q3

The sum of the following series

$$1 + 6 + \frac{9(1^2 + 2^2 + 3^2)}{7} + \frac{12(1^2 + 2^2 + 3^2 + 4^2)}{9} + \frac{15(1^2 + 2^2 + \ldots + 5^2)}{11} + \ldots \text{ up to 15 terms, is:}$$

(1) 7520 (2) 7510

(3) 7830 (4) 7820

9 Jan Evening

Q4

Let a, b and c be the 7^{th}, 11^{th} and 13^{th} terms respectively of a non-constant A.P. If these are also the three consecutive terms of a G.P., then $\frac{a}{c}$ is equal to:

(1) 2 (2) $\frac{1}{2}$

(3) $\frac{7}{13}$ (4) 4

9 Jan Evening

Q5

Let $a_1, a_2, a_3, \ldots, a_{10}$ be in G.P. with $a_i > 0$ for $i = 1, 2, \ldots, 10$ and S be the set of pairs (r, k), $r, k \in \mathbb{N}$ (the set of natural numbers) for which

$$\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$$

Then the number of elements in S, is :

(1) 4 (2) infinitely many

(3) 2 (4) 10

10 Jan Evening

Q6

The sum of an infinite geometric series with positive terms is 3 and the sum of the cubes of its terms is $\frac{27}{19}$. Then the common ratio of this series is :

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$

(3) $\frac{2}{9}$ (4) $\frac{4}{9}$

11 Jan Morning

Q7

Let $a_1, a_2, ..., a_{10}$ be a G.P. If $\frac{a_3}{a_1} = 25$, then $\frac{a_9}{a_5}$ equals :

(1) 5^4 (2) $4(5^2)$
(3) 5^3 (4) $2(5^2)$

11 Jan Morning

Q8

Let α and β be the roots of the quadratic equation $x^2 \sin\theta - x(\sin\theta\cos\theta + 1) + \cos\theta = 0$ $(0 < \theta < 45°)$, and $\alpha < \beta$. Then $\sum_{n=0}^{\infty}\left(\alpha^n + \frac{(-1)^n}{\beta^n}\right)$ is equal to :

(1) $\frac{1}{1-\cos\theta} - \frac{1}{1+\sin\theta}$

(2) $\frac{1}{1+\cos\theta} + \frac{1}{1-\sin\theta}$

(3) $\frac{1}{1-\cos\theta} + \frac{1}{1+\sin\theta}$

(4) $\frac{1}{1+\cos\theta} - \frac{1}{1-\sin\theta}$

11 Jan Evening

Q9

If 19th term of a non-zero A.P. is zero, then its (49th term) : (29th term) is :

(1) 4 : 1 (2) 1 : 3
(3) 3 : 1 (4) 2 : 1

11 Jan Evening

Q10

The product of three consecutive terms of a G.P. is 512. If 4 is added to each of the first and the second of these terms, the three terms now form an A.P. Then the sum of the original three terms of the given G.P. is :

(1) 36 (2) 32
(3) 24 (4) 28

12 Jan Morning

Q11

Let $S_k = \frac{1+2+3+.....+k}{2}$.

If $S_1^2 + S_2^2 + ... + S_{10}^2 = \frac{5}{12}A$, Then A is equal to

(1) 283 (2) 301
(3) 303 (4) 156

12 Jan Morning

Q12

If ${}^nC_4$, ${}^nC_5$ and ${}^nC_6$ are in A.P., then n can be :

(1) 9 (2) 14
(3) 11 (4) 12

12 Jan Evening

Q13

If the sum of the first 15 terms of the series

$\left(\frac{3}{4}\right)^3 + \left(1\frac{1}{2}\right)^3 + \left(2\frac{1}{4}\right)^3 + 3^3 + \left(3\frac{3}{4}\right)^3 + ...$ is equal to 225 k then k is equal to :

(1) 108 (2) 27
(3) 54 (4) 9

12 Jan Evening

Q14

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

(1) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (2) d, e, f are in A.P.

(3) d, e, f are in G.P. (4) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

8 April Evening

Q15

The sum $\sum_{k=1}^{20} k \frac{1}{2^k}$ is equal to :

(1) $2 - \frac{3}{2^{17}}$ (2) $1 - \frac{11}{2^{20}}$

(3) $2 - \frac{11}{2^{19}}$ (4) $2 - \frac{21}{2^{20}}$

8 April Evening

Q16

Let the sum of the first n terms of a non-constant A.P., a_1, a_2, a_3, ………….. be $50n + \frac{n(n-7)}{2}A$, where A is a constant. If d is the common difference of this A.P., then the ordered pair (d, a_{50}) is equal to:

(1) (50, 50 + 46A) (2) (50, 50 + 45A)

(3) (A, 50 + 45A) (4) (A, 50 + 46A)

9 April Morning

Q17

If the sum and product of the first three terms in an A.P. are 33 and 1155, respectively, then a value of its 11^{th} term is:

(1) –35 (2) 25 (3) –36 (4) –25

9 April Evening

Q18

The sum of the series $1 + 2 \times 3 + 3 \times 5 + 4 \times 7 + \ldots$.. upto 11^{th} term is:

(1) 915 (2) 946 (3) 945 (4) 916

9 April Evening

Q19

If $a_1, a_2, a_3, \ldots.. a_n$ are in A.P. and $a_1 + a_4 + a_7 + \ldots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to :

(1) 98 (2) 76 (3) 38 (4) 64

10 April Morning

Q20

The sum $\frac{3\times1^3}{1^2} + \frac{5\times(1^3+2^3)}{1^2+2^2} + \frac{7\times(1^3+2^3+3^3)}{1^2+2^2+3^2}) + \ldots\ldots$ upto 10^{th} term, is :

(1) 680 (2) 600 (3) 660 (4) 620

10 April Morning

Q21

The sum $1 + \frac{1^3+2^3}{1+2} + \frac{1^3+2^3+3^3}{1+2+3} + \ldots\ldots +$ $\frac{1^3+2^3+3^3+\ldots+15^3}{1+2+3+\ldots+15} - \frac{1}{2}(1+2+3+\ldots+15$ is equal to :

(1) 620 (2) 1240 (3) 1860 (4) 660

10 April Evening

Q22

Let a, b and c be in G.P. with common ratio r, where $a \uparrow 0$ and $0 < r'' \frac{1}{2}$. If $3a$, $7b$ and $15c$ are the first three terms of an A.P., then the 4^{th} term of this A.P. is :

(1) $\frac{2}{3}a$ (2) $5a$ (3) $\frac{7}{3}a$ (4) a

10 April Evening

Q23

For $x \in R$, let $[x]$ denote the greatest integer $\le x$, then the sum of the series

$$\left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - \frac{1}{100}\right] + \left[-\frac{1}{3} - \frac{2}{100}\right] + \cdots + \left[-\frac{1}{3} - \frac{99}{100}\right]$$ is

(1) –153 (2) –133 (3) –131 (4) –135

12 April Morning

Q24

If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\alpha^r + \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\beta^r$ is equal to :

(1) $\frac{21}{346}$ (2) $\frac{29}{358}$ (3) $\frac{1}{12}$ (4) $\frac{7}{116}$

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Q25

Let S_n denote the sum of the first n terms of an A.P. If $S_4 = 16$ and $S_6 = -48$, then S_{10} is equal to :

(1) –260 (2) –410 (3) –320 (4) –380

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Q26

If $a_1, a_2, a_3, \ldots\ldots$ are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is :

(1) 200 (2) 280 (3) 120 (4) 150

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Q27

If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to :

(1) 0 (2) $\alpha\beta$ (3) $\alpha\gamma$ (4) $\beta\gamma$

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JEE Mains 2019 Chapter wise Question Bank

Sequences and Series - Solutions

Q1

(4) $\because$ a, b, c, are in G.P.

$\Rightarrow \quad b^2 = ac$

Since, $a + b + c = xb$

$\Rightarrow \quad a + c = (x - 1)b$

Take square on both sides, we get

$a^2 + c^2 + 2ac = (x - 1)^2 b^2$

$\Rightarrow \quad a^2 + c^2 = (x - 1)^2 ac - 2ac \qquad [\because b^2 = ac]$

$\Rightarrow \quad a^2 + c^2 = ac[(x - 1)^2 - 2]$

$\Rightarrow \quad a^2 + c^2 = ac[x^2 - 2x - 1]$

$\because$ $a^2 + c^2$ are positive and $b^2 = ac$ which is also positive.

Then, $x^2 - 2x - 1$ would be positive but for $x = 2$, $x^2 - 2x - 1$ is negative.

Hence, x cannot be taken as 2.

9 Jan Morning

Q2

(1) $$S = \sum_{i=1}^{30} a_i = \frac{30}{2}[2a_1 + 29d]$$

$$T = \sum_{i=1}^{15} a_{(2i-1)} = \frac{15}{2}[2a_1 + 28d]$$

Since, $S - 2T = 75$

$\Rightarrow 30a_1 + 435d - 30a_1 - 420d = 75$

$\Rightarrow d = 5$

Also, $a_5 = 27 \quad \Rightarrow \quad a_1 + 4d = 27$

$\Rightarrow a_1 = 7$,

Hence, $a_{10} = a_1 + 9d = 7 + 9 \times 5 = 52$

9 Jan Morning

Q3

(4) $$S = 1 + 6 + \frac{9(1^2+2^2+3^2)}{7} + \frac{12(1^2+2^2+3^2+4^2)}{9}$$

$$+ \frac{15(1^2+2^2+3^2+4^2+5^2)}{11} + \ldots$$

$$S = \frac{3\cdot(1)^2}{3} + \frac{6\cdot(1^2+2^2)}{5} + \frac{9\cdot(1^2+2^2+3^2)}{7} + \frac{12\cdot(1^2+2^2+3^2+4^2)}{9} + \ldots$$

Now, n^{th} term of the series,

$$t_n = \frac{3n\cdot(1^2+2^2+\ldots+n^2)}{(2n+1)}$$

$$\Rightarrow \quad t_n = \frac{3n\cdot n(n+1)(2n+1)}{6(2n+1)} = \frac{n^3+n^2}{2}$$

$$\therefore \quad S_n = \Sigma t_n = \frac{1}{2}\left\{\left(\frac{n(n+1)}{2}\right)^2 + \frac{n(n+1)(2n+1)}{6}\right\}$$

$$= \frac{n(n+1)}{4}\left(\frac{n(n+1)}{2} + \frac{2n+1}{3}\right)$$

Hence, sum of the series upto 15 terms is,

$$S_{15} = \frac{15\times16}{4}\left\{\frac{15\cdot16}{2} + \frac{31}{3}\right\}$$

$$= 60 \times 120 + 60 \times \frac{31}{3}$$

$$= 7200 + 620$$

$$= 7820$$

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Q4

(4) Let first term and common difference be A and D respectively.

$\therefore\ a = A + 6D,\ b = A + 10D$

and $c = A + 12D$

Since, a, b, c are in G.P.

Hence, relation between a, b and c is,

$\therefore\ b^2 = a.c.$

$\therefore\ (A + 10D)^2 = (A + 6D)\ (A + 12D)$

$\therefore\ 14D + A = 0$

$\therefore\ A = -\ 14D$

$\therefore\ a = -8D,\ b = -4D$ and $c = -2D$

$\therefore\ \dfrac{a}{c} = \dfrac{-8D}{-2D} = 4$

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Q5

(2) Let common ratio of G.P. be R

$\Rightarrow\quad a_2 = a_1R,\ a_3 = a_1R^2, \ldots\ a^{10} = a_1R^9$

$C_1 \to C_1 - C_2,\ C_2 \to C_2 - C_3$

$$\Delta = \begin{vmatrix} \ln\left(\dfrac{a_1^r a_2^k}{a_2^r a_3^k}\right) & \ln\left(\dfrac{a_2^r a_3^k}{a_3^r a_4^k}\right) & \ln a_3^r a_4^k \\ \ln\left(\dfrac{a_4^r a_5^k}{a_5^r a_6^k}\right) & \ln\left(\dfrac{a_5^r a_6^k}{a_6^r a_7^k}\right) & \ln a_6^r a_7^k \\ \ln \dfrac{a_7^r a_8^k}{a_8^r a_9^k} & \ln\left(\dfrac{a_8^r a_9^k}{a_9^r a_{10}^k}\right) & \ln a_9^r a_{10}^k \end{vmatrix}$$

$$\Delta = \begin{vmatrix} \ln \dfrac{1}{R^{r+k}} & \ln \dfrac{1}{R^{r+k}} & \ln a_3^r a_4^k \\ \ln \dfrac{1}{R^{r+k}} & \ln \dfrac{1}{R^{r+k}} & \ln a_6^r a_7^k \\ \ln \dfrac{1}{R^{r+k}} & \ln \dfrac{1}{R^{r+k}} & \ln a_9^r a_{10}^k \end{vmatrix} = 0$$

$\forall r, K \in N$

Hence, number of elements in S is infinitely many.

10 Jan Evening

Q6

(2) Let the terms of infinite series are $a, ar, ar^2, ar^3, \ldots$

So, $\dfrac{a}{1-r} = 3$

Since, sum of cubes of its terms is $\dfrac{27}{19}$ that is sum of a^3, $a^3r^3, \ldots \infty$ is $\dfrac{27}{19}$.

So, $\dfrac{a^3}{1-r^3} = \dfrac{27}{19}$

$\Rightarrow\quad \dfrac{a}{1-r} \times \dfrac{a^2}{(1+r^2+r)} = \dfrac{27}{19}$

$\Rightarrow\quad \dfrac{9(1+r^2-2r)\times 3}{1+r^2+r} = \dfrac{27}{19}$

$\Rightarrow\quad 6r^2 - 13r + 6 = 0$

$\Rightarrow\quad (3r-2)(2r-3) = 0$

$\Rightarrow\quad r = \dfrac{2}{3}$, or $\dfrac{3}{2}$

As $|r| < 1$

So, $r = \dfrac{2}{3}$

11 Jan Morning

Q7

(1) Let $a_1 = a,\ a_2 = ar,\ a_3 = ar^2 \ldots a_{10} = ar^9$ where r = common ratio of given G.P.

Given, $\dfrac{a_3}{a_1} = 25$

$\Rightarrow\quad \dfrac{ar^2}{a} = 25$

$\Rightarrow\quad r = \pm 5$

Now, $\text{—} = \dfrac{ar^8}{ar^4} = r^4 = (\pm 5)^4 = 5^4$

11 Jan Morning

Q8

(3) $x^2 \sin\theta - x(\sin\theta \cdot \cos\theta + 1) + \cos\theta = 0.$

$x^2 \sin\theta - x \sin\theta \cdot \cos\theta - x + \cos\theta = 0.$

$x \sin\theta - (x - \cos\theta) - 1(x - \cos\theta) = 0.$

$(x - \cos\theta)(x \sin\theta - 1) = 0.$

$\therefore \quad x = \cos\theta, \operatorname{cosec}\theta, \theta \in (0, 45°)$

$\therefore \quad \alpha = \cos\theta, \beta = \operatorname{cosec}\theta$

$$\sum_{n=0}^{\infty} \alpha^n = 1 + \cos\theta + \cos^2\theta + \ldots \infty = \frac{1}{1-\cos\theta}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\beta^n} = 1 - \frac{1}{\operatorname{cosec}\theta} + \frac{1}{\operatorname{cosec}^2\theta} - \frac{1}{\operatorname{cosec}^3\theta} + \ldots \infty$$

$$= 1 - \sin\theta + \sin^2\theta - \sin^3\theta + \ldots \infty.$$

$$= \frac{1}{1+\sin\theta}$$

$$\therefore \quad \sum_{n=0}^{\infty}\left(\alpha^n + \frac{(-1)^n}{\beta^n}\right) = \sum_{n=0}^{\infty} \alpha^n + \sum_{n=0}^{\infty} \frac{(-1)^n}{\beta^n}$$

$$= \frac{1}{1-\cos\theta} + \frac{1}{1+\sin\theta}.$$

11 Jan Evening

Q9

(3) Let first term and common difference of AP be a and d respectively, then

$$t_n = a + (n-1)d$$

$$\therefore \quad t_{19} = a + 18d = 0$$

$$\therefore \quad a = -18d \qquad \ldots(1)$$

$$\therefore \quad \frac{t_{49}}{t_{29}} = \frac{a+48d}{a+28d}$$

$$= \frac{-18d+48d}{-18d+28d} = \frac{30d}{10d} = 3$$

$t_{49} : t_{29} = 3 : 1$

11 Jan Evening

Q10

(4) Let three terms of a G.P. be $\frac{a}{r}, a, ar$

$$\frac{a}{r} \cdot a \cdot ar = 512$$

$$a^3 = 512$$

$$a = 8$$

4 is added to each of the first and the second of three terms then three terms are, $\frac{8}{r} + 4, 8 + 4, 8r.$

$\because \frac{8}{r} + 4, 12, 8r$ form an A.P.

$$\therefore \quad 2 \times 12 = \frac{8}{r} + 8r + 4$$

$$\Rightarrow \quad 2r^2 - 5r + 2 = 0$$

$$\Rightarrow \quad (2r-1)(r-2) = 0$$

$$\Rightarrow \quad r = \frac{1}{2} \quad \text{or} \quad 2$$

Therefore, sum of three terms $= \frac{8}{2} + 8 + 16 = 28$

12 Jan Morning

Q11

(3) $\because\ 1+2+3+\ldots+k=\frac{k(k+1)}{2}$

$$\therefore S_k=\frac{k(k+1)}{2k}=\frac{k+1}{2}$$

$$\Rightarrow \quad \frac{5}{12}A=\frac{1}{4}[2^2+3^3+\ldots+11^2]$$

$$=\frac{1}{4}[1^2+2^2+\ldots+11^2-1]$$

$$=\frac{1}{4}\left[\frac{11(11+1)(2\times 11+1)}{6}-1\right]$$

$$\frac{1}{4}\left[\frac{11\times 12\times 23}{6}-1\right]$$

$$=\frac{1}{4}[505]$$

$$A=\frac{505}{4}\times\frac{12}{5}=303$$

12 Jan Morning

Q12

(2) Since ${}^nC_4$, ${}^nC_5$ and ${}^nC_6$ are in A.P.

$$2\,{}^nC_5={}^nC_4+{}^nC_6$$

$$2=\frac{{}^nC_4}{{}^nC_5}+\frac{{}^nC_6}{{}^nC_5}$$

$$2=\frac{5}{n-4}+\frac{n-5}{5}$$

$$\Rightarrow \quad 12(n-4)=30+n^2-9n+20$$

$$\Rightarrow \quad n^2-21n+98=0$$

$$(n-7)(n-14)=0$$

$$(n-7)(n-14)=0$$

$$n=7,\ n=14$$

12 Jan Evening

Q13

(2) $$S=\left(\frac{3}{4}\right)^3+\left(\frac{3}{2}\right)^3+\left(\frac{9}{4}\right)^3+(3)^3+\ldots$$

$$S=\left(\frac{3}{4}\right)^3+\left(\frac{6}{4}\right)^3+\left(\frac{9}{4}\right)^3+\left(\frac{12}{4}\right)^3+\ldots$$

Let the general term of S be

$$T_r=\left(\frac{3r}{4}\right)^3\text{, then}$$

$$255K=\sum_{r=1}^{15}T_r=\left(\frac{3}{4}\right)^3\sum_{r=1}^{15}r^3$$

$$255K=\frac{27}{64}\times\left(\frac{15\times 16}{2}\right)^2$$

$$\Rightarrow \quad K=27$$

12 Jan Evening

Q14

(1) Since a, b, c are in G.P.

$\therefore\ b^2=ac$

Given equation is, $ax^2+2bx+c=0$

$$\Rightarrow ax^2+2\sqrt{ac}x+c=0 \Rightarrow (\sqrt{a}x+\sqrt{c})^2=0$$

$$\Rightarrow x=-\sqrt{\frac{c}{a}}$$

Also, given that $ax^2+2bx+c=0$ and $dx^2+2ex+f=0$ have a common root.

$$\Rightarrow x=-\sqrt{\frac{c}{a}}\ \text{ must satisfy } dx^2+2ex+f=0$$

$$\Rightarrow d.\frac{c}{a}+2e\left(-\sqrt{\frac{c}{a}}\right)+f=0$$

$$\frac{d}{a}-\frac{2e}{\sqrt{ac}}+\frac{f}{c}=0 \quad\Rightarrow\quad \frac{d}{a}-\frac{2e}{b}+\frac{f}{c}=0$$

$$\Rightarrow \frac{2e}{b}=\frac{d}{a}+\frac{f}{c} \Rightarrow \frac{d}{a},\frac{e}{b},\frac{f}{c}\ \text{ are in A.P.}$$

8 April Evening

Q15

(3) Let, $S=\sum_{k=1}^{20} k.\frac{1}{2^k}$

$$S=\frac{1}{2}+2.\frac{1}{2^2}+3.\frac{1}{2^3}+....+20.\frac{1}{2^{20}} \quad ...(i)$$

$$\frac{1}{2}S=\frac{1}{2^2}+2.\frac{1}{2^3}+....+19\frac{1}{2^{20}}+20\frac{1}{2^{21}} \quad ...(ii)$$

On subtracting equations (ii) by (i),

$$\frac{S}{2}=\left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+....+\frac{1}{2^{20}}\right)-20\frac{1}{2^{21}}$$

$$=\frac{\frac{1}{2}\left(1-\frac{1}{2^{20}}\right)}{1-\frac{1}{2}}-20.\frac{1}{2^{21}}=1-\frac{1}{2^{20}}-10.\frac{1}{2^{20}}$$

$$\frac{S}{2}=1-11.\frac{1}{2^{20}} \Rightarrow S=2-11.\frac{1}{2^{19}}=2-\frac{11}{2^{19}}$$

8 April Evening

Q16

(4) $\because S_n=\left(50-\frac{7A}{2}\right)n+n^2\times\frac{A}{2} \quad \Rightarrow \ a_1=50-3S$

$\therefore\ d=a_2-a_1=\left(S_{n_2}-S_{n_1}\right)-S_{n_1}$

$\Rightarrow d=\frac{A}{2}\times 2=A$

Now, $a_{50}=a_1+49\times d$

$=(50-3A)+49A$

$=50+46A$

So, $(d, a_{50})=(A, 50+46A)$

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Q17

(4) Let three terms of A.P. are $a-d, a, a+d$

Sum of terms is, $a-d+a+a+d=33 \Rightarrow\ a=11$

Product of terms is, $(a-d)\,a\,(a+d)=11(121-d^2)=1155$

$\Rightarrow 121-d^2=105 \Rightarrow d=\pm 4$

if $d=4$

$T_{11}=T_1+10d=7+10(4)=47$

if $d=-4$

$T_{11}=T_1+10d=15+10(-4)=-25$

9 April Evening

Q18

(2) $1+2.3+3.5+4.7+........$

Let, $S=(2.3+3.5+4.7+.......)$

Now, $S_{10}=\sum_{n=1}^{10}(n+1)(2n+1)=\sum_{n=1}^{10}\left(2n^2+3n+1\right)$

$$=\frac{2n(n+1)(2n+1)}{6}+\frac{3n(n+1)}{2}+n$$

Put $n=10$

$$=\frac{2.10.11.21}{6}+\frac{3.10.11}{2}+10=945$$

Hence required sum of the series $=1+945=946$

9 April Evening

Q19

(2) $a_1+a_4+a_7+....+a_{16}=114$

$\Rightarrow 3(a_1+a_{16})=114$

$\Rightarrow a_1+a_{16}=38$

Now, $a_1+a_6+a_{11}+a_{16}=2(a_1+a_{16})=2\times 38=76$

10 April Morning

Q20

(3) r^{th} term of the series, $T_r = \dfrac{(2r+1)(1^3+2^3+3^3+...+r^3)}{1^2+2^2+3^2+...+r^2}$

$$T_r = (2r+1)\left(\frac{r(r+1)}{2}\right)^2 \times \frac{6}{r(r+1)(2r+1)} = \frac{3r(r+1)}{2}$$

$$\therefore \text{ sum of 10 terms is } = S = \sum_{r=1}^{10} T_r = \frac{3}{2}\sum_{r=1}^{10}(r^2+r)$$

$$= \frac{3}{2}\left\{\frac{10\times(10+1)(2\times10+1)}{6} + \frac{10\times11}{2}\right\}$$

$$= \frac{3}{2}\times5\times11\times8 = 660$$

10 April Morning

Q21

(1) Let, $S = 1 + \dfrac{1^3+2^3}{1+2} + \dfrac{1^3+2^3+3^3}{1+2+3} + ...15 \text{ terms}$

$$T_n = \frac{1^3+2^3+...n^3}{1+2+...n} = \frac{\left(\frac{n(n+1)}{2}\right)^2}{\frac{n(n+1)}{2}} = \frac{n(n+1)}{2}$$

$$\text{Now, } S = \frac{1}{2}\left(\sum_{n=1}^{15} n^2 + \sum_{n=1}^{15} n\right) = \frac{1}{2}\left(\frac{15(16)(31)}{6} + \frac{15(16)}{2}\right)$$

$$= 680$$

$$\therefore \text{ required sum is, } 680 - \frac{1}{2}\frac{15(16)}{2} = 680 - 60 = 620$$

10 April Evening

Q22

(4) $\because$ a, b, c are in G.P. $\Rightarrow b = ar, c = ar^2$

$\because$ $3a, 7b, 15c$ are in A.P. $\Rightarrow 3a, 7ar, 15ar^2$ are in A.P.

$\therefore$ $14ar = 3a + 15ar^2$

$$\Rightarrow 15r^2 - 14r + 3 = 0 \Rightarrow r = \frac{1}{3} \text{ or } \frac{3}{5}$$

$$\because r < \frac{1}{2} \quad \therefore r = \frac{3}{5} \text{ rejected}$$

Fourth term $= 15ar^2 + 7ar - 3a$

$$= a(15r^2 + 7r - 3) = a\left(\frac{15}{9} + \frac{7}{3} - 3\right) = a$$

10 April Evening

Q23

(2) $\because [x] + \left[x + \dfrac{1}{n}\right] + \left[x + \dfrac{2}{n}\right] .... \left[x + \dfrac{n-1}{n}\right] = [nx]$

and $[x] + [-x] = -1$ $(x \notin z)$

$$\therefore \quad \left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - 100\right] + ... + \left[-\frac{1}{3} - \frac{99}{100}\right]$$

$$= -100 - \left\{\left[\frac{1}{3}\right] + \left[\frac{1}{3} + \frac{1}{100}\right] + ... \left[\frac{1}{3} + \frac{99}{100}\right]\right\}$$

$$= -100 - \left[\frac{100}{3}\right] = -133$$

12 April Morning

Q24

(3) Given equation is, $375x^2 - 25x - 2 = 0$

Sum and product of the roots are,

$$\alpha + \beta = \frac{25}{375} \text{ and } \alpha\beta = \frac{-2}{375}$$

$$\lim_{n\to\infty} \sum_{r=1}^{n} (\alpha^r + \beta^r)$$

$$= (\alpha + \alpha^2 + \alpha^3 + ...\infty(\beta + \beta^2 + \beta^3 ... + \infty)$$

$$= \frac{\alpha}{1-\alpha} + \frac{\beta}{1-\beta} = \frac{\alpha+\beta-2\alpha\beta}{1-(\alpha+\beta)+\alpha\beta}$$

$$= \frac{\frac{25}{375} + \frac{4}{375}}{1 - \frac{25}{375} - \frac{2}{375}} = \frac{29}{375-25-2} = \frac{29}{348} = \frac{1}{12}$$

12 April Morning

Q25

(3) Given, $S_4 = 16$ and $S_6 = -48$

$\Rightarrow$ $2(2a + 3d) = 16 \Rightarrow 2a + 3d = 8$ ...(i)

And $3[2a + 5d] = -48 \Rightarrow 2a + 5d = -16$

$\Rightarrow$ $2d = -24$ [using equation (i)]

$\Rightarrow$ $d = -12$ and $a = 22$

$$\therefore \quad S_{10} = \frac{10}{2}(44 + 9(-12)) = -320$$

12 April Morning

Q26

(1) Let the common difference of the A.P. is 'd'.

Given, $a_1 + a_7 + a_{16} = 40$

$\Rightarrow \quad a_1 + a_1 + 6d + a_1 + 15d = 40$

$\Rightarrow \quad 3a_1 + 21d = 40$

$\Rightarrow \quad a_1 + 7d = \frac{40}{3}$ ...(i)

Now, sum of first 15 terms of this A.P. is,

$S_{15} = \frac{15}{2}[2a_1 + 14d] = 15(a_1 + 7d)$

$= 15\left(\frac{40}{3}\right) = 200$ [Using (i)]

12 April Evening

Q27

(4) $\because$ α, β, γ are three consecutive terms of a non-constant G.P.

$\therefore \quad \beta^2 = \alpha\gamma$

So roots of the equation $\alpha x^2 + 2\beta x + \gamma = 0$ are

$\frac{-2\beta \pm 2\sqrt{\beta^2 - \alpha\gamma}}{2\alpha} = \frac{\beta}{\alpha}$

$\because \quad \alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root.

$\therefore$ this root satisfy the equation $x^2 + x - 1 = 0$

$\beta^2 - \alpha\beta - \alpha^2 = 0$

$\Rightarrow \quad \alpha\gamma - \alpha\beta - \alpha^2 = 0 \Rightarrow \alpha + \beta = \gamma$

Now, $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$

$= \alpha\beta + \beta^2 = (\alpha + \beta)\beta = \beta\gamma$

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